

- **Remote diagnostics and maintenance:** IoE allows manufacturers to remotely diagnose vehicle issues and schedule maintenance, improving safety and convenience.
- **Autonomous vehicles:** IoE plays a crucial role in enabling autonomous vehicles to perceive their surroundings, communicate with other vehicles, and navigate safely.

Personal safety and security

- **Smart homes and buildings:** IoE can equip homes and buildings with real-time security systems, including intrusion detection, access control, and fire alarms.
- **Wearable safety devices:** Devices like smartwatches with fall detection can send alerts to emergency services in real-time, potentially saving lives.
- **Personal tracking devices:** Real-time tracking of individuals, particularly children or elderly individuals prone to wandering, can ensure their safety and well-being.

Entertainment and gaming

- **Smart TVs and streaming devices:** IoE allows for personalised recommendations and real-time content updates.
- **Interactive gaming:** It enables immersive gaming experiences where players interact with the environment in real-time.
- **Connected fitness devices:** IoE fitness trackers and equipment can provide real-time feedback.

IoT security principles

We now look at the top 8 IoT security principles that have been laid down keeping modern security threats and vulnerabilities in mind.

No universal passwords

Concept: Moving away from a single password for all devices. Each device should have a unique and strong password or use alternative

authentication methods like biometrics (fingerprint scan) or multi-factor authentication (MFA).

Real-time example: Imagine a home with multiple smart devices, where each device, like the smart lock, thermostat, and camera, has its own unique password or requires fingerprint verification for access. This prevents a single compromised password from granting access to all devices.

Secured interfaces

Concept: Securing communication channels between devices, gateways, and the cloud using strong encryption protocols like TLS/SSL. This protects data from eavesdropping and tampering during transmission.

Real-time example: A fitness tracker transmits workout data to a smartphone app. Data encryption ensures only authorised devices can access the data, safeguarding sensitive health information.

Proven cryptography

Concept: Implementing established and well-tested cryptographic algorithms for encryption and decryption of data. This ensures the confidentiality, integrity, and authenticity of data throughout its life cycle.

Real-time example: A smart home hub uses AES-256 encryption to secure communication between connected devices and the hub. This industry-standard algorithm makes it incredibly difficult for attackers to decipher the data.

Security by default

Concept: Devices should be shipped with security features enabled by default, requiring no manual configuration by users. This simplifies security for non-technical users and ensures a baseline level of protection.

Real-time example: A new smart speaker comes pre-configured with strong password requirements and automatic software update functionality. This eliminates the risk of users leaving the device vulnerable with weak

passwords or outdated software.

Signed software updates

Concept: Updates should be digitally signed by the manufacturer to verify their authenticity and prevent installing malicious firmware disguised as updates.

Real-time example: A notification appears on a smart TV prompting for a software update. The update is digitally signed by the manufacturer, assuring users it's legitimate and not a tampered version.

Software updates applied automatically

Concept: Devices should automatically download and install security patches whenever they become available. This ensures devices are always protected against the latest vulnerabilities.

Real-time example: A smart thermostat automatically downloads and installs a security patch that addresses a recently discovered vulnerability in its software. This eliminates the need for manual intervention and ensures timely protection.

Vulnerability reporting scheme

Concept: Establishing a clear and accessible channel for users and researchers to report discovered vulnerabilities to manufacturers. This allows for prompt identification, investigation, and patching of vulnerabilities.

Real-time example: Security researchers discover a vulnerability in a popular smart home camera brand. They report the vulnerability through the manufacturer's dedicated vulnerability reporting portal. The manufacturer promptly investigates, develops a patch, and releases it to all users.

Security expiration date

Concept: Assigning a life span to the security capabilities of a device. Manufacturers should clearly communicate when a device will no longer receive security updates or support, allowing users to make